

WHEN FUSION WEIGHTS AND REPRESENTATION UTILITY DISAGREE IN MOS PREDICTION

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ABSTRACT

Automatic Mean Opinion Score (MOS) prediction offers a scalable alternative to costly listening tests, and modern predictors often combine different kinds of speech representations. Understanding which representations matter for MOS prediction can guide model design and explain what predictors rely on. A common shortcut is to read the learned fusion weights as importance scores. However, a fusion weight only reflects the role of a representation within one fitted combination, and it may not capture its broader usefulness. In this work, we show that fusion weights and representation utility often disagree. Representations that received almost no weight still improved predictions in other combinations, mainly when a stronger substitute was missing. Yet, a richer attribution method did not select better compact subsets for unseen systems than simple rankings. These results suggest that fusion weights offer only a partial view of which representations matter for MOS prediction.

Index Terms— mean opinion score, speech quality, representation analysis, Shapley value, model interpretation

1. INTRODUCTION

Subjective listening tests remain the reference standard for assessing synthetic and processed speech, but they are costly and hard to scale. Non-intrusive Mean Opinion Score (MOS) prediction offers an automatic alternative by learning to reproduce human quality ratings. Early approaches relied on task-specific architectures [1], while recent systems build on pretrained self-supervised speech representations [2]. MOS predictors have combined several encoders or feature types [3, 4, 5]. Challenge reports also document a broader range of evaluation domains and approaches [6, 7]. These developments make it important to understand which representations contribute to predicted quality.

A common way to judge importance is to look at the learned fusion weights. These weights, however, are fitted for one specific set of representations. A representation may receive almost no weight when a similar one is present, yet still be useful when that substitute is missing. This matters in practice. Each retained representation adds its own extraction pipeline at inference time, while discarding one that looks unimportant may hurt performance on new synthesis systems. Explaining a fused model and choosing which representations to keep are therefore related but different problems, and neither has been studied systematically for MOS prediction.

In this work, we study which representations matter for MOS prediction and whether fusion weights give a reliable answer. We

compare fusion weights with representation utility across combinations of speech representations for MOS prediction. We use ten feature extractors on four datasets: BRSpeechMOS [8], BVCC [9], SingMOS [10], and TMHINT-QI [11]. We first compare the fusion weights of the full model with three other views of importance, namely how well each representation performs alone, how much the model loses without it, and how much it helps on average across all combinations. We then test whether rankings based on each of these views select small subsets that generalize to unseen systems. To keep this test fair, the evaluation systems never take part in training or selection.

The main contributions are as follows:

- We propose an audit of MOS predictors that separates the fusion weight of each representation from its standalone performance, its leave-one-out effect, and its average contribution across combinations.
- Across four corpora, we find 16 of 40 corpus and representation pairs with near-zero fusion weight but positive Shapley utility, and we examine how stable this pattern is across refits and alternative groupings of the feature extractors.
- We separate explaining representations from selecting them. Under nested system-disjoint evaluation, Shapley utility reveals redundancy and substitution between representations, but it gives no significant advantage over simply ranking representations by standalone performance once multiple comparisons are corrected for.

Our results show that fusion weights give an incomplete explanation of MOS predictors. The representations that received almost no weight yet remained useful were mostly simple acoustic and pitch features, while none of the large pretrained speech encoders showed this behavior. Removing these features from the full model barely changed its error, but they still helped when other representations were missing. This suggests that their information about quality is not absent but overlaps with what other representations already provide. However, this richer explanation did not select better compact subsets for unseen systems. A method that is good at explaining a model is therefore not necessarily good at choosing which representations to keep.

2. RELATED WORK

Early MOS predictors such as MOSNet relied on task-specific architectures trained from scratch [1]. Later work showed that pre-

trained self-supervised speech representations provide strong features for this task [2], and many systems now combine several of them. Yang et al. [3] fused multiple wav2vec 2.0, HuBERT, and WavLM models. SAMOS [4] integrated complementary semantic and acoustic representations. Other studies compared supervised, self-supervised, and speaker-verification representations [8], or selected SSL layers to improve accuracy and reduce cost [12]. These works study fusion, representation comparisons, and layer selection. They do not establish whether a weight learned in one fused model measures usefulness in other combinations.

Outside MOS prediction, research on model interpretability has warned against treating learned weights as general importance scores. Attention weights, for example, do not always reflect which inputs drive a prediction [13]. Shapley values offer an alternative by averaging the contribution of each input over all possible combinations of the other inputs [14], and SAGE extends this idea to measure how much each feature contributes to overall model performance [15]. Feature importance depends on the chosen value function and the features already present; Shapley-based explanations also have methodological limitations [15, 16]. Attribution also differs from selection. A Shapley value describes average contribution, but it does not directly identify the best small subset of inputs.

Any study of representation importance also depends on how predictors are evaluated. Generalizing to unseen synthesis systems is a central goal of automatic MOS prediction, yet strong results within a benchmark do not always carry over to new systems or listening tests [17, 9]. When a corpus contains multiple utterances per system, random utterance-level splits can put a system in both training and evaluation. This may give optimistic estimates for unseen systems, motivating system-grouped evaluation [17, 18].

3. METHODS

We use the official training, validation, and test partitions of BR-SpeechMOS, BVCC, SingMOS, and TMHINT-QI, modeling each corpus independently. Their training partitions contain 6, 175, 31, and 98 systems, respectively. Audio is converted to mono at 16 kHz, and no representation is adapted using MOS labels.

The ten feature extractors were chosen to provide complementary views of speech. Whisper [19] provides representations shaped by speech recognition, while ContentVec [20] emphasizes linguistic content with reduced speaker variation. WavLM [21] provides a broader self-supervised representation that can retain content, speaker, and paralinguistic information. BEATs [22] and CED [23], trained for general audio representation and tagging, add information about acoustic events and sound characteristics. An equivalent rectangular bandwidth (ERB) frontend [24] captures spectral energy on an auditory-motivated frequency scale, whereas an ECAPA-TDNN encoder [25] targets speaker characteristics. Finally, eGeMAPS [26] provides interpretable prosodic, voice-quality, and spectral descriptors. To examine pitch-related prosody at two levels, we use RMVPE, originally proposed for vocal pitch estimation in polyphonic music [27], to derive continuous F0 summary statistics and a quantized pitch-histogram representation. We use Whisper large-v3, WavLM Base+, BEATs iter3+, CED-Small, and eGeMAPSv02; learned sequence representations are mean-pooled. Exact checkpoints, feature-extraction parameters, and preprocessing configurations are provided in the supplementary material.

We fixed the ridge penalty at $\alpha = 10$ after fold-wise standardization. This common value was chosen as a controlled reference rather than optimized separately for each representation or coalition, preventing regularization differences from being interpreted as repre-

sentation utility. A grouped sensitivity sweep over $\alpha \in [10^{-2}, 10^7]$ changed only 4 of 40 selection cells, with test-MSE differences no larger than 0.019; the qualitative conclusions were therefore stable over a broad regularization range.

We use five system-grouped folds to obtain out-of-fold (OOF) predictions: each ridge expert predicts systems excluded from its own preprocessing and fitting. For each non-empty group of experts, we then choose a single set of non-negative weights that sum to one and use it to average their OOF predictions. The experts keep their intercepts, and the average adds none. After estimating these weights from training data, we refit the experts on the full training partition but keep the weights fixed for validation and test. Thus, neither validation nor test labels influence the fusion weights; the validation-to-test audit evaluates a fixed fusion rule.

With $v(S) = -\mathcal{L}(S)$ and the empty coalition predicting the training MOS mean, exact Shapley utility is

$$\phi_j = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [v(S \cup \{j\}) - v(S)]. \quad (1)$$

We evaluate all $2^{10} - 1 = 1,023$ non-empty coalitions using utterance-level MSE. A discordant case has full-gate weight $w_j < 10^{-6}$ and positive point-estimate Shapley utility. Validation selects the cases before frozen test evaluation; test labels do not fit experts, gates, or preprocessing. Test intervals in the attribution audit are system-bootstrap summaries and are descriptive and unadjusted. For comparison, leave-one-out utility is $\Delta_j^{\text{LOO}} = v(N) - v(N \setminus \{j\})$. It conditions only on the full set, whereas Eq. (1) averages over every coalition size. A positive Shapley value paired with negligible Δ_j^{LOO} is therefore the operational signature of contextual substitutability examined here; it is not a test of whether the encoder contains unique or causal information.

To check whether the coalition results depend on the fusion rule, we recompute utilities from the same expert predictions using three alternatives to the simplex gate. Uniform averaging tests the coalition ranking without fitting weights. Non-negative least squares with an intercept tests whether that ranking persists when weights may be zero but need not sum to one, following the use of non-negative coefficients in stacked regression [28]. Ridge stacking provides a further check with regularized, potentially signed coefficients. These comparisons assess the robustness of coalition utilities; claims about zero simplex weights apply only to the simplex gate.

We use nested, system-grouped cross-validation to select subsets and evaluate them on different systems [29, 18]. In each of five outer folds, four inner folds refit preprocessing and experts, calibrate gates, compute gate, singleton, and Shapley rankings, and search all subsets with $k \in 1, 2, 3, 5$. We then refit the selected experts on the outer-development systems and evaluate the fixed subset and gate once on the held-out systems. The objective at both levels averages the mean squared error across systems, giving each system equal weight.

Confirmatory inference uses 10,000 paired bootstrap draws of held-out systems within each corpus, shared across selectors and budgets. We normalize cell differences by the sampled full-model MSE and average them over 16 corpus-budget cells. Holm correction covers five prespecified contrasts: gate, singleton, and Shapley versus exhaustive search, and gate and Shapley versus singleton [30]. Exploratory BRSpeechMOS-excluded and leave-one-corpus-out analyses reaggregate saved outer-held errors without refitting and repeat the paired bootstrap and contrasts on retained corpora. They assess sensitivity to corpus composition, not generalization to new corpora.

4. RESULTS AND DISCUSSION

Allocation–utility mismatch. The validation audit identifies 16/40 pairs with $w_j < 10^{-6}$ and $\phi_j > 0$. Figure 1(a) places these cases in the context of all 40 pairs, and panel (b) shows their frozen test estimates. Fifteen of 16 retain positive test point estimates, and 11/16 have descriptive unadjusted 95% system-bootstrap intervals whose lower bounds exceed zero. The count is unchanged when the numerical-zero threshold varies from 10^{-8} to 10^{-3} , although exact zeros remain solver-tolerance dependent. Five test effects are below 2% of full-model MSE, whereas BRSpeechMOS Speaker/CED and BVCC/TMHINT-QI eGeMAPS exceed 17%. BRSpeechMOS estimates are especially uncertain because its training partition has only six systems.

All 16 cases have absolute leave-one-out utility below 1.1×10^{-7} MSE. Once all competitors are present, removing the zero-weight expert therefore changes loss negligibly; averaged over other coalitions, however, the same expert can contribute when substitutes are absent. This is evidence of contextual substitutability within the audited linear-expert/simplex-gate system. It does not show that the representation contains unique acoustic information, should be deployed, or would receive the same attribution under a different architecture.

There is no optimization paradox in this outcome. With frozen expert predictions, full-gate fitting minimizes a convex quadratic over one simplex; a boundary solution $w_j^{(N)} = 0$ states only that reallocating mass to expert j does not improve that objective. It imposes no equality between $v(S \cup j)$ and $v(S)$ for the 2^9 predecessor coalitions entering ϕ_j . Strongly correlated prediction columns can therefore yield a stable mixture prediction but non-unique or interchangeable allocations. This conditional nature of the coefficient is what the complete-game audit exposes.

Fusion-rule sensitivity (exploratory). NNLS reproduces the mismatch in 8/9 eligible cases outside BRSpeechMOS but 0/4 within it. Uniform averaging and ridge stacking change coalition utilities but do not define a comparable zero-simplex-weight event. Thus, the NNLS analogue depends on the corpus, and the other fusion rules serve as robustness checks rather than independent confirmations of the primary claim.

Feature-extractor grouping and refit robustness (exploratory). Because Shapley utility is conditional on how the feature extractors are grouped, we also tested whether the allocation–utility mismatch persists after merging or removing overlapping prosodic views. The original game with ten feature extractors yielded 16/40 discordant corpus–representation pairs. Grouping the two RMVPE views yielded 13/36 cases, grouping all prosodic views yielded 9/32, and removing both quantized RMVPE and eGeMAPS yielded 7/32. Thus, the exact incidence depends on the representation game, but the qualitative mismatch is not specific to the original feature-extractor grouping.

The contextual rankings were considerably more stable than the complete-gate support. Across the alternative feature-extractor groupings, grouped-Shapley rank correlations with the original game ranged from 0.976 to 1.000. In separate system-level refits, the four corpora produced 51, 4, 17, and 14 distinct complete-gate supports, respectively. On the frozen attribution cases, 8/16 retained positive Shapley utility in all 50 training-system refits. Complete tables and refit distributions are provided in the supplementary material.

These diagnostics reinforce the distinction between allocation and contextual utility. A particular zero coefficient is conditional on the fitted competitor set and may be unstable across refits, whereas the broader contextual ordering can remain comparatively stable. At

Table 1. Confirmatory macro held-system contrasts relative to exhaustive inner-development selection. Values average four corpora and $k \in \{1, 2, 3, 5\}$; negative values indicate lower equal-system MSE. CIs use 10,000 paired system bootstraps, and p -values are Holm-adjusted across five prespecified contrasts. Percentages are relative to full-model MSE.

Selector	Scores	Δ MSE [95% CI]	p_{Holm}
Gate	1	−1.51 [−2.82, +0.74]	.694
Singleton	10	−0.72 [−2.91, +1.44]	1.000
Shapley	1,023	−2.15 [−3.77, +0.71]	.694

the same time, the reduction in discordant cases under alternative feature-extractor groupings shows that the observed rate is not an intrinsic property of an encoder. We therefore interpret the mismatch as a property of the specified predictive game rather than as representation-level importance.

Selection on unseen systems (confirmatory). Table 1 summarizes the three ranking rules relative to exhaustive inner-development selection. None differs after Holm correction. The Shapley-minus-singleton contrast is −1.43% (95% CI [−3.23, −0.07]%; raw $p = .0296$), but its adjusted value is $p_{\text{Holm}} = .148$; failure to reject does not establish equivalence. Exact Shapley uses 1,023 coalition scores versus ten singleton scores, without a detectable multiplicity-adjusted selection benefit here. The exhaustive reference is itself selected on finite inner-development data, so a ranking method can obtain a negative outer contrast by generalizing better than the inner optimum. The reported intervals consequently compare selection procedures rather than claiming that exhaustive search is an oracle. Likewise, a non-significant adjusted contrast is neither evidence of equivalence nor a bound on effects outside the four evaluated corpora.

Corpus sensitivity (exploratory). Because the largest cell deviations arise from BRSpeechMOS, we reaggregate the saved outer-held errors without refitting. Excluding BRSpeechMOS changes the Shapley-minus-exhaustive point estimate from −2.15% to +0.63%. Leave-one-corpus-out reaggregation likewise shows that the macro summary depends on corpus composition. These are post-hoc robustness analyses, not new evidence of cross-corpus generalization, and they do not change the prespecified four-corpus Holm family. The sign reversal is therefore evidence about fragility of the aggregate point estimate, not evidence that Shapley is worse after removing BRSpeechMOS. Full leave-one-corpus-out intervals and adjusted values should be interpreted as a small set of dependent sensitivity summaries because every reaggregation shares three corpora and the same saved outer-fold predictions.

Inferential status. The validation rule and frozen test evaluation were specified for the primary allocation–utility audit, but its 16 test intervals are descriptive and are not a familywise test. Confirmatory inference is confined to the five macro held-system contrasts and their joint Holm correction in Table 1. The alternative-fusion, BRSpeechMOS-excluded, and leave-one-corpus-out analyses were added as exploratory robustness checks. Their role is to identify dependence on the aggregation rule and corpus composition, not to enlarge the confirmatory claim.

Scope and limitations. These findings concern frozen, mean-pooled representations, ridge experts, and a global simplex gate; other architectures or utility functions may behave differently. The attribution audit uses official partitions and descriptive, unadjusted test intervals. Nested evaluation tests unseen systems within each corpus, not

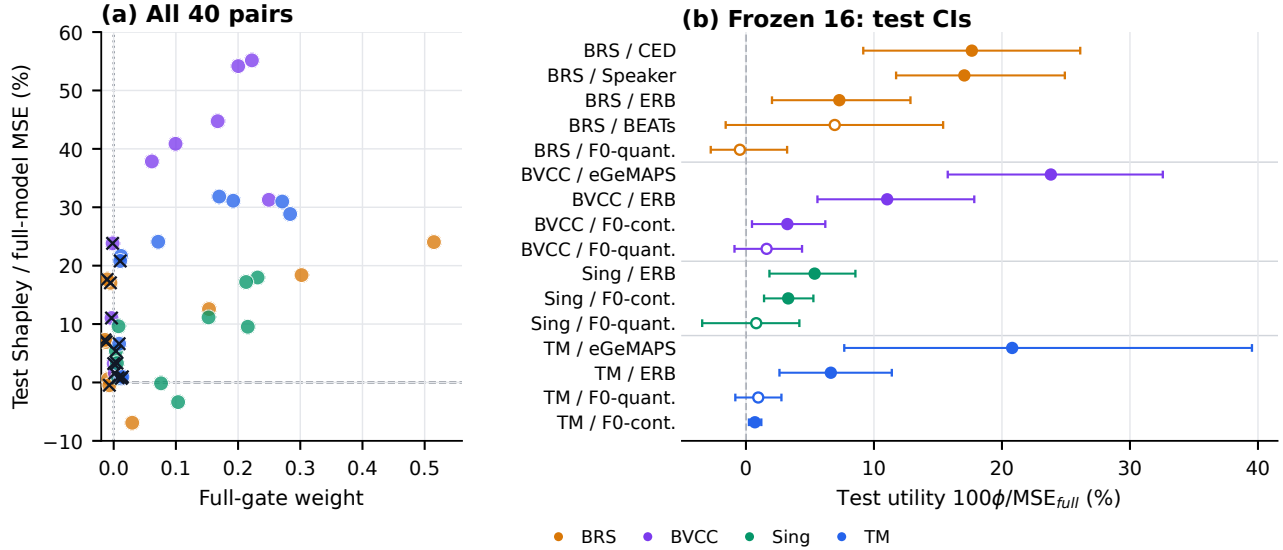


Fig. 1. Allocation and contextual utility under the primary frozen-ridge, simplex-gate model. (a) All 40 corpus–representation pairs: frozen full-gate weight versus test Shapley utility normalized by full-model MSE. Crosses mark the 16 cases selected on validation by $w_j < 10^{-6}$ and positive validation utility; colors denote corpora. (b) Frozen test estimates and descriptive, unadjusted 95% system-bootstrap intervals for those 16 cases. Filled markers have interval lower bounds above zero; open markers have intervals that include zero. The audit uses utterance-level MSE and makes no multiplicity-adjusted claim for the 16 intervals.

generalization to new corpora. BRSSpeechMOS has only six training systems, making its sensitivity estimates uncertain. Shapley values depend on feature grouping, and correlated experts can yield unstable allocations; numerical zeros also depend on solver tolerance. Coalition counts do not measure extraction cost.

5. CONCLUSION

This study shows that fusion weights should not be interpreted as general measures of representation importance in MOS prediction. Within the evaluated frozen-expert and simplex-gated setting, 16 of 40 corpus–representation pairs received negligible full-mixture weight while retaining positive Shapley utility, with 15 remaining positive by test point estimate. Together with negligible leave-one-out effects, this indicates contextual substitutability: a representation may be unnecessary when all alternatives are present yet remain predictive when substitutes are absent. Although this mismatch varies across corpora, fusion rules, and feature-extractor groupings, its persistence shows that mixture allocation and contextual utility answer different questions. This distinction also matters for representation selection. Despite evaluation of the complete coalition lattice, exact Shapley attribution showed no Holm-adjusted advantage over simpler ranking criteria for selecting fixed-budget subsets on unseen systems. Shapley attribution can diagnose redundancy and substitution, but should not be assumed to yield better compact models. For deployment, subsets should be selected and evaluated directly under system-disjoint conditions and explicit computational constraints. Overall, fusion coefficients explain how a fitted model distributes predictive responsibility—not which representations are intrinsically useful or which subset should be deployed.

Reproducibility statement. Upon acceptance, the implementation, configurations, analysis scripts, and supporting materials will be released at [mos-fusion-weight-audit](https://mos-fusion-weight-audit.github.io).

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